

ABET Self-Study Evaluation and Continuous Improvement Report

Course: CHE 312 – Transport Phenomena II, Spring 2025

1. Self-Study Evaluation (Findings and Interpretation)

The assessment of Student Outcome 1 (SO1) in CHE 312 was conducted using the final exam as the primary direct assessment instrument. The exam was scored out of 100 points and targeted students' ability to apply heat and mass transfer principles in engineering problem-solving contexts.

The results indicate **low attainment of expected learning outcomes with substantial variability among students**. The **average exam score of 39.25** and **standard deviation of 19.95** reflect both weak overall performance and significant dispersion in student achievement. This suggests that many students are not consistently mastering foundational concepts such as heat conduction, convection correlations, diffusion, and coupled transport processes.

The **performance indicator average of 1.85 (on a 3-point rubric scale)** falls **below the program benchmark of 2.0**, indicating that **SO1 is not attained** for this course offering. The variability in rubric scores (standard deviation = 0.81) further highlights inconsistent student mastery across the cohort.

The distribution of student performance reinforces these findings:

- **Satisfactory (S): 5 students**
- **Developing (D): 7 students**
- **Unsatisfactory (U): 8 students**

With **75% of students not achieving satisfactory performance**, the results indicate widespread difficulty in applying engineering principles to complex heat and mass transfer problems. Only a small subset of students demonstrated adequate proficiency in solving multi-step transport phenomena problems involving formulation, analysis, and interpretation.

Observed student weaknesses include:

- Difficulty formulating governing equations for heat and mass transfer systems
- Inconsistent application of boundary conditions and physical assumptions
- Limited ability to integrate coupled transport mechanisms
- Weak interpretation of physical meaning behind mathematical solutions

Overall, the findings suggest that students are not adequately prepared for the cognitive and analytical demands of advanced transport phenomena.

2. Continuous Improvement Plan

To address the deficiencies identified in SO1 attainment, the following targeted continuous improvement actions will be implemented:

A. Strengthening Foundational Knowledge

- Reinforce prerequisite concepts from thermodynamics, fluid mechanics, and applied mathematics at the beginning of the course.
- Provide structured review materials and short diagnostic assessments early in the semester to identify knowledge gaps.

B. Increasing Structured Practice and Formative Feedback

- Implement **weekly low-stakes quizzes and guided homework assignments** focused on core heat and mass transfer principles.
- Provide **rapid, detailed feedback** to correct misconceptions before summative assessments.
- Use formative assessment results to adjust pacing and instructional emphasis.

C. Enhancing Problem-Solving Instruction

- Introduce **explicit step-by-step problem-solving frameworks** for conduction, convection, and diffusion problems.
- Emphasize:
 - Physical system definition
 - Governing equation formulation
 - Assumption justification
 - Solution interpretation
- Increase exposure to **worked examples of exam-level complexity** with instructor explanation of reasoning processes.

D. Expanding Active Learning and Student Engagement

- Incorporate **in-class collaborative problem-solving activities** and peer instruction.
- Use conceptual questions to reinforce understanding of heat and mass transfer mechanisms.
- Encourage students to explain reasoning verbally and in written form during problem-solving exercises.

E. Strengthening Academic Support Structures

- Provide **targeted tutoring and structured review sessions**, especially prior to exams.
- Identify at-risk students early using quiz/homework performance indicators.
- Encourage increased participation in office hours and supplemental instruction sessions.

F. Improving Assessment Alignment

- Conduct a detailed review of **alignment between learning outcomes, instructional delivery, and exam difficulty level**.
- Perform item-level analysis to identify specific topics with highest failure rates (e.g., convection correlations, diffusion equations).
- Refine rubric descriptors to improve clarity, consistency, and discriminatory power between performance levels.

3. Reassessment Plan

The effectiveness of these improvements will be evaluated in the next offering of CHE 312 by tracking:

- Increase in **average rubric scores above the 2.0 attainment threshold**
- Reduction in the proportion of **Unsatisfactory (U) performance levels**
- Improvement in **overall exam average and reduction in score variability**
- Enhanced student performance on key heat and mass transfer topics through item analysis

Findings will be documented and used to determine whether additional pedagogical or curricular adjustments are required.

4. Conclusion

The assessment indicates that **Student Outcome 1 is not attained in CHE 312** for this cycle. The results demonstrate significant gaps in student understanding and application of heat and mass transfer principles, along with high variability in performance. The continuous improvement plan focuses on strengthening foundational preparation, increasing structured practice and feedback, enhancing instructional methods, and improving assessment alignment. These actions are expected to improve student learning outcomes and achieve consistent attainment in future course offerings.